

Hybrid Biomedical Image-Analysis Pipeline: A Modular Case Study

Dennis Mburu Nganga

24179558

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Modality: fluorescence microscopy (nuclei) · Local VLM: qwen2.5vl:7b via Ollama (CPU)

1 Introduction and data

This project builds an auditable nuclei-analysis pipeline that replaces informal diagnostic impressions with a deterministic chain: raw image → segmentation mask → region-property quantification → structured JSON → narrative. The dataset is synthetic fluorescence microscopy at native 256×256 (80 train / 20 validation / 12 test) supplied with binary ground-truth masks, 16-bit instance labels, and per-image true object counts in metadata.csv. That count column is the reason every claim below is quantified rather than asserted; no resizing was required.

Every stage ran end to end: validation macro Dice 0.9961, 100% schema-valid JSON, and 100% agreement between the LLM-emitted object count and the computed one. The pipeline is nonetheless not trustworthy. It undercounts dense and clustered fields by 43–67%, its density classification is right about half the time, and its quality flag returned "ok" on every corrupted image it was shown. This report is mostly about why those three failures survive a 0.9961 Dice.

Grayscale strategy. The nuclear signal is almost entirely in the blue channel. ITU-R 601-2 luma conversion (0.299R + 0.587G + 0.114B) evaluates to roughly 0.355-I here, discarding about two thirds of the dynamic range. Extracting the blue channel directly is one line of code executed before any segmentation decision, and it nearly triples contrast:

Metric	Luma conversion	Blue-channel extraction	Impact
Contrast (std)	17.20	48.49	2.82× higher contrast
Peak value	90	255	Full 8-bit range retained
Global Otsu threshold	34.5	99.2	Cleaner background separation

Pooled histograms confirmed the bimodal dark-field structure that Otsu assumes (Fig. 1, column 2).

2 Task 1 — VLM prompt engineering

The brief's suggested llama3.2-vision could not be loaded (mllama architecture failure in the installed Ollama binary), so qwen2.5vl:7b was substituted. Two prompts were compared: a naive "Describe this image in detail", and a structured prompt fixing four keys (modality, tissue_type, notable_features, image_quality), permitting "uncertain", forbidding prose outside the JSON, and anchoring the role to description rather than diagnosis.

The naive prompt averaged 752 characters of unparseable free text at ~96 s per call; the structured prompt returned 195-character schema-valid JSON in ~21 s. More important than speed is drift. Pairwise Jaccard similarity across three identical naive calls was 0.307, meaning roughly 69% of words changed between runs. All three agreed on the modality, so the instability is not in the central claim but in the volunteered detail: run 2 asserted the image was "likely taken using a confocal microscope", an instrument claim with no visual basis, and a downstream record built on that run would carry it as fact. The structured prompt cut length standard deviation to exactly 0.0, but its Jaccard was only 0.532 with zero identical pairs — the free-text notable_features field still drifted. **Reproducibility follows the individual constraint, not the prompt as a whole.**

A temperature sweep was inconclusive and slightly perverse: at T = 0.0, image_quality returned "moderate" once and "good" twice; at T = 0.8, all three runs returned "good". With n = 3 this is noise, but it does show that zero temperature does not guarantee identical VLM output.

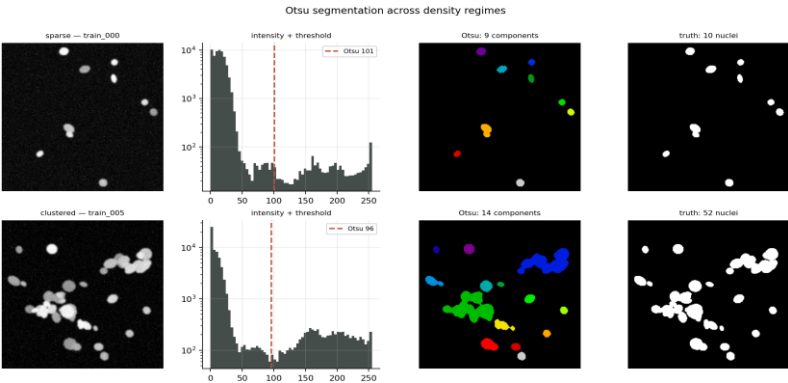
3 Task 2 — Otsu segmentation and classical features

Images were thresholded with global Otsu, cleaned morphologically (opening before closing, so specks are removed before larger objects are sealed), labelled by connected components, and quantified with regionprops_table (area, eccentricity, solidity, mean intensity). Comparing Otsu component counts against the true counts in metadata.csv exposes where the error actually lives:

Density regime	Otsu count error	Perfect-mask ceiling	Attributable to Otsu (pp)
Sparse	-3.2%	-2.7%	-0.5
Normal	-19.5%	-14.9%	-4.6
Dense	-46.0%	-38.1%	-7.9
Clustered	-62.5%	-57.8%	-4.7

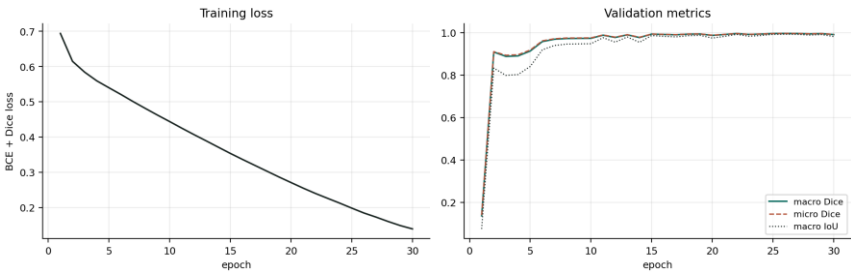
Figure 1 — Otsu at the two density extremes: source, histogram with threshold, labelled components, ground truth. The clustered row recovers 14 components from 52 nuclei despite a clearly bimodal histogram.

Counting connected components in a *perfect* ground-truth mask still undercounts clustered nuclei by 57.8%. Otsu contributes only 4.7 percentage points on top of that. **The error therefore belongs to the task formulation — binary segmentation cannot separate touching objects — not to the thresholding method.** No amount of model sophistication removes a ceiling that is present in the label itself, which is the single most consequential finding in this report and the reason the U-Net result in §4 must be read sceptically.



Task 3 — U-Net segmentation and loss ablation

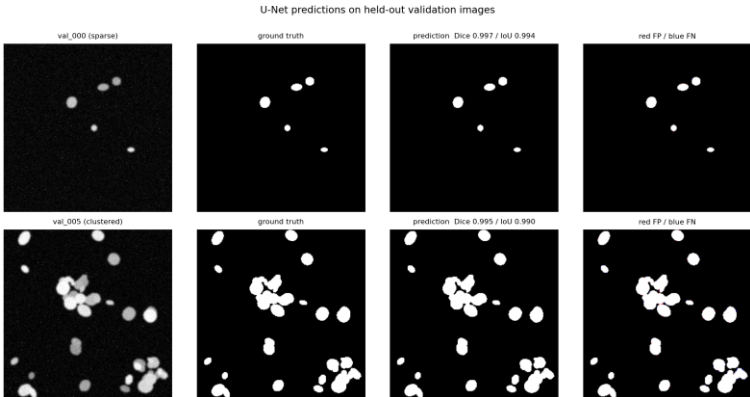
Figure 2 — Training/validation loss and validation Dice. The Dice curve is flat from ~epoch 15 onward, which is what makes the loss ablation below uninformative. The provided U-Net skeleton is deliberately narrow (base = 16, three encoder levels, ~483k parameters) and was trained from scratch on the 80 training images. Augmentation was restricted to exact, label-preserving transforms (flips and 90° rotations), which avoids interpolating the binary mask at the cost of never exposing the model to intermediate orientations. Validation macro Dice rose from 0.137 at epoch 1 to a plateau near 0.99 by epoch 15, with the best checkpoint at epoch 26. Training loss continued to fall past that point while validation stayed on the plateau — consistent with the model fitting training-set specifics that do not transfer, but since validation never meaningfully degraded this is saturation, not classical overfitting.



Model / loss	Macro Dice	Macro IoU	Notes
Otsu + morphology (baseline)	0.9733	—	no training; U-Net wins on all 20 val images
U-Net — Dice loss	0.9961	0.9923	best checkpoint; per-image 0.9935–0.9977
U-Net — BCE + Dice	0.9961	—	equal epoch budget
U-Net — BCE only	0.9955	—	equal epoch budget

Micro Dice/IoU for the Dice-loss model were 0.9960 / 0.9921, agreeing with the macro figures to within 0.0001, so uneven object density across images is not skewing the metric. Per-image standard deviation is ≈0.001, which makes the 0.0006 spread across the three losses unresolvable: the task is near-saturated on pixel overlap, so the loss function has no room to matter. Reporting a "best loss" here would be reporting noise. Mask agreement between U-Net and Otsu is ≈0.95 IoU — the two methods largely find the same pixels and differ mainly at boundaries.

Pixel superiority masks a counting failure. On the test set the U-Net reported 43 objects against 75 true for test_004 (–43%), 36 against 78 for test_010 (–54%), and 16 against 49 for test_005 (–67%). Figure 3 — U-Net input, ground truth, prediction, error map (red = FP, blue = FN). Dice 0.997 sparse / 0.995 clustered; the clustered case merges nuclei in both the prediction and the label. Figure 3 shows why the two facts coexist: the error maps are nearly empty, with false predictions confined to a boundary rim one or two pixels wide, and in the clustered row the prediction and the ground truth merge *the same* touching nuclei. Dice measures pixel overlap and is blind to whether a contiguous region contains one nucleus or five (Reinke et al., 2021). A model can reach 0.9961 Dice while undercounting by 43% precisely because the mask it is matching carries the same binary defect. For a task whose scientific output is a count, Dice is the wrong headline metric.



Task 4 — Hybrid pipeline and structured records

The full chain (raw image → U-Net mask → regionprops table → LLM structured JSON → LLM narrative) ran across all 12 unseen test images with 100% parse success, 100% schema validity, and 100% of n_objects values copied exactly from the computed table.

```
{ "image_id": "test_000", "segmenter": "unet", "n_objects": 8, "mean_area": 192.0,
  "coverage_pct": 2.34, "mean_eccentricity": 0.631, "mean_solidity": 0.962,
```

```
"density_class": "sparse", "shape_regularity": "regular", "quality_flag": "ok",
"narrative": "The image contains 8 distinct objects with a mean area of 192.0 px, a
median area of 179.5 px, and a standard deviation of 103.77 px.",
"llm_n_objects": 8, "n_objects_matches_computed": true, "parse_status": "clean" }
```

Excerpt, test_010 (78 true nuclei): "n_objects": 36 — "n_objects_matches_computed": true — "quality_flag": "ok" — "parse_status": "clean".

Structural fidelity was perfect and semantic classification failed anyway. density_class was correct on 40% of validation and 50% of test images, and across all 12 test images the LLM emitted only "sparse" or "normal" — never "dense" or "clustered", even where those regimes were present. The causal chain is strictly upstream. test_010 contains 78 nuclei, but the binary formulation ceiling meant segmentation isolated only 36 objects; to a model reading a numerical table, 36 objects in a 256×256 field genuinely reads as sparse, and that is what it returned. The classification is a correct inference over corrupted input, not a hallucination. JSON-as-source-of-truth therefore delivered perfect number fidelity and zero protection against the numbers being wrong: constraining what a language model may *invent* does nothing about what the pipeline *measures*.

6 Extension — robustness to corruption

Four provided corrupted images (from two sources) were traced through all four stages: image statistics → mask → feature table → narrative.

Image	Mask IoU vs clean	Count shift	Quality flag
test_000_blur	0.484	8 → 7	ok
test_000_lowcontrast	0.023	8 → 1	ok
test_004_blur	0.485	43 → 9	ok
test_004_lowcontrast	0.216	43 → 1	ok

Corruption is detectable at stage 1, but only under a standard-deviation criterion (25.8 → 3.9 for low contrast, → 14.4 for blur). A mean-based check catches low contrast and misses blur entirely, which moves the mean by 0.5 grey levels while halving the object count. **Blur is the more dangerous corruption precisely because it is less visible:** low contrast produces an obviously broken mask (one component, ~100% coverage) that any sanity check or human reviewer spots immediately, whereas test_004_blur shifts 43 objects to 9 at IoU 0.485 — and nine nuclei is an entirely believable measurement. Degradation that preserves plausibility is how confident wrong data enters a record.

Figure 4 — Corruption propagation, test_000. Bottom row: U-Net masks (8, 7, 1 components). The low-contrast mask, outlined for visibility, covers essentially the whole field — yet quality_flag stayed "ok". **The guard rail failed by construction.** quality_flag was keyed to trigger on zero objects or on a large area standard deviation; because low contrast merged the field into exactly one object, the area standard deviation was zero and both triggers were evaded. The underlying cause is that U-Net input is scaled by a fixed constant (± 255) and never normalised per image, so an affine intensity shift moves the input outside the distribution in which its decision boundary was learned. Robustness here is not a property of the architecture; it is a property of whether the decision rules adapt per image.

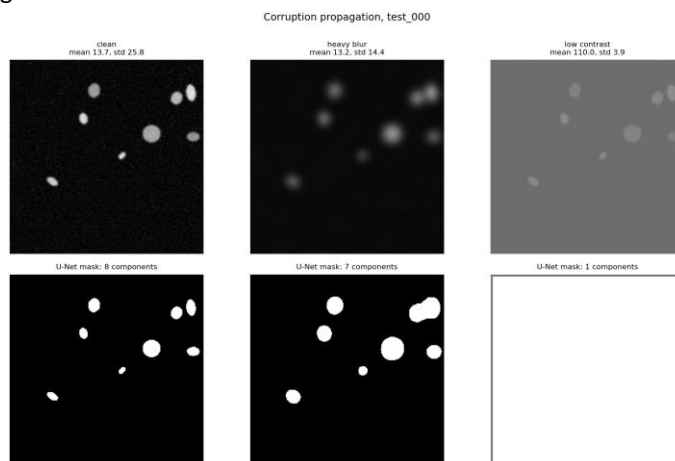
7 Critical questions

1. Which description is more useful, and which more trustworthy? The VLM description is more useful for high-level semantic context — it identifies fluorescence microscopy against a dark field with no rule-writing — but the numbers-first description is far more trustworthy. The VLM cannot count or measure sub-structures, so its claims are unverifiable and, as §2 showed, it volunteers different unsupported specifics on each call. The numbers-first route abstracts the image into auditable metrics first, so every sentence is traceable to a value in the feature table.

2. Did the U-Net improve on Otsu? On pixel overlap, universally: it beat Otsu on all 20 validation images with no counter-example (narrowest margin val_011, 0.996 vs 0.979), and it separated clustered nuclei better — in train_005 (52 true) Otsu merged touching cells for a -73% count error. Otsu was adequate on ultra-sparse fields such as train_006 (5 true) at a fraction of the cost, so on sparse data the training overhead buys nothing.

3. Report Dice and IoU. What do they mean, and where does the model err? Macro Dice 0.9961, macro IoU 0.9923. Dice is the harmonic overlap of predicted and true foreground pixels ($2|A \cap B| / (|A| + |B|)$); IoU is $|A \cap B| / |A \cup B|$ and is always the harsher of the two, which is why 0.9961 Dice corresponds to 0.9923 IoU. Both are pixel-set measures and neither is object-aware. Errors are confined almost entirely to boundary rims one or two pixels wide (Fig. 3) and to junctions between touching nuclei, where the model merges objects — the same merge the ground-truth mask makes, which is exactly why the metric does not penalise it. The high value therefore reports that the model reproduces the label, not that it counts nuclei correctly.

4. Where can the LLM hallucinate, and what reduces the risk? There are three exposed points. In Task 1 the VLM invents visual specifics outright (the "confocal microscope" claim). In Task 2/4 the LLM could restate the numeric fields incorrectly when producing the record. And in the narrative step it could introduce clinical or causal language the features do not support. The mitigations were a closed output schema, closed vocabularies on classification fields, a permitted "uncertain" value, and a programmatic check (n_objects_matches_computed) that re-derives the count from the feature table rather



than trusting the model — which held at 100%. Keeping the JSON as source of truth means the prose is a rendering of the record rather than its origin, so drift in the narrative never corrupts the numbers. It does not, as §5 showed, make the numbers right.

5. Would you trust this clinically, and what one change helps most? No. The data is synthetic, the training set is 80 images, the validation split is 20, and §6 demonstrates catastrophic sensitivity to a corruption the system rated "ok". The single highest-value change is per-image intensity normalisation before segmentation, which removes the affine sensitivity that caused the low-contrast collapse at essentially zero cost. It does not, however, lift the counting ceiling; that needs an instance-aware formulation such as StarDist (Schmidt et al., 2018), which predicts object shapes rather than a binary mask and so can separate touching nuclei.

8 Limitations and conclusion

The dataset is synthetic and lacks staining artefacts, focal drift, debris and real biological variability, so the near-saturated Dice is a property of the data more than of the model. Sample sizes are small throughout (20 validation images; $n = 3\text{--}5$ for the variability and temperature runs), so several comparisons are directional rather than significant. The dominant limitation is structural: binary segmentation cannot separate touching objects, and every downstream number — count, density class, narrative — inherits that ceiling. The pipeline is complete, auditable and internally consistent, and confidently wrong about the one quantity it exists to produce. That combination, not the Dice score, is the result worth reporting.

References

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